Rishikesh Ombase

Agentic AI Architect · Senior Data Scientist

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SUMMARY

Agentic AI architect and Senior Data Scientist with 6+ years building machine learning and LLM systems, mostly in finance. Designs agentic systems end to end: **multi-agent orchestration** with LangGraph, **MCP tool layers**, agentic RAG and memory, model routing across frontier and open-weight models, and the **evals, guardrails and observability** that make agents safe in production. Works from a layered reference architecture and keeps humans in the loop on anything irreversible.

CORE SKILLS

Agentic AI	LangGraph, CrewAI, OpenAI Agents SDK, Claude Agent SDK, multi-agent systems, MCP servers, A2A, computer use, human-in-the-loop
Models	Claude, GPT, Gemini, Llama, Qwen, DeepSeek, vision-language models, fine-tuning (LoRA, QLoRA, DPO), model routing
Platform & cloud	Python, FastAPI, Pydantic, Docker, Kubernetes, vLLM, Ollama, Redis, PostgreSQL, AWS Bedrock, Azure AI Foundry, Vertex AI
Knowledge & memory	RAG, GraphRAG, hybrid search and reranking, LlamaIndex, Qdrant, pgvector, Neo4j, agent memory, context engineering
LLMOps & governance	Ragas, DeepEval, Langfuse, LangSmith, guardrails, prompt-injection defence, OpenTelemetry
Machine learning	PyTorch, scikit-learn, time-series forecasting, classification, computer vision (OpenCV), A/B testing

EXPERIENCE

Senior Data Scientist · AIT Global India
Dec 2023 – Present

- Designed and built LLM-integrated API services and automated pipelines that extract insights from financial reports using Python, OCR and SQL.
- Developed generative AI modules for financial commentary summarisation and sentiment analysis.
- Collaborated on risk-scoring models with AI-augmented forecasting capabilities.
- Designed and deployed the APIs behind portfolio intelligence dashboards.

AI Data Specialist · Futurense Technologies
Jul 2023 – Oct 2023

- Built real-time web data collection that enriched datasets for pricing models and market-trend analysis.
- Delivered executive decision dashboards with Python, Power BI and SQL.

Software Engineer · Systech Solutions
Jul 2022 – Mar 2023

- Designed data frameworks for large-scale trading datasets on AWS.
- Enabled real-time updates to the analytics dashboards that monitor asset performance.

Data Scientist · Varilligence Technologies
Jan 2020 – Mar 2022

- Researched and built machine learning models for stock-price movement prediction using PySpark and financial indicators.
- Built alpha-generating trading signals from historical equities data, macro indicators and sentiment variables.
- Contributed to model validation and A/B testing of quant strategies in simulation environments.

SELECTED PROJECTS

AlphaDesk · Multi-agent quant research platform In progress

LangGraph · MCP · GraphRAG · Qdrant · Neo4j · code sandbox · Langfuse · Kubernetes

- Hierarchical supervisor agent that delegates to market-data (MCP servers), research (GraphRAG over filings and news), quant (sandboxed forecasting and backtests) and risk agents.
- Model routing between frontier and open-weight models; every run traced in Langfuse and gated by an eval suite; human approval required before any signal leaves the system.

FundDocs Studio · Agentic styling for fund filings

Vision-language models · LLM agents · LangGraph · style-guide RAG · structured outputs

- Multi-agent document pipeline that takes prospectuses, summary prospectuses and SAI documents together with the client's style guide.
- Vision-language models read each page's layout, a style-guide agent decides which rule applies to every heading, table and footnote, and LLM agents apply the formatting, verify it against the guide and route documents for human sign-off.

ListingStudio · Multimodal real-estate listing generator

Vision LLM · GPT models · FastAPI · prompt engineering · structured outputs · image classification

- Classified property images by room and feature, then generated listing content at scale with LLMs: views and surroundings, amenities, neighbourhood details and full descriptions.
- Added a critic step that checks generated claims against the photos and listing data before publishing.

CineMind · OTT recommendation engine

FastAPI · PostgreSQL · Redis · embeddings · clustering · Docker · microservices

- Grouped viewers into taste clusters from watch history so users with similar habits receive similar films; candidates from the cluster and item similarity are ranked per user.
- Redis serves precomputed recommendations, sessions and trending lists; FastAPI microservices on PostgreSQL with authentication, media data management and access control.

Invoice Data Automation · Document AI

Python · OCR · NLP · Selenium · email parsing · scheduling

- Automated extraction and cleaning of invoice data from Word and PDF documents with NLP and OCR, then filled target systems automatically, improving speed and accuracy.

Earlier work: drone-type recognition with computer vision on Apache Spark · FOX Channel data integration for the Looker BI team · automated mutual-fund NAV pipeline · quantitative stock-price forecasting.

EDUCATION, RECOGNITION & CERTIFICATIONS

M.Sc. Computer Science
Shivaji University, Kolhapur · 2022

B.Sc. Computer Science
Mahavir College, Kolhapur · 2019

Best Performer of the Month
Spot Award
GOM Award

Machine Learning in Python
Data Analysis with Python
Data Visualization